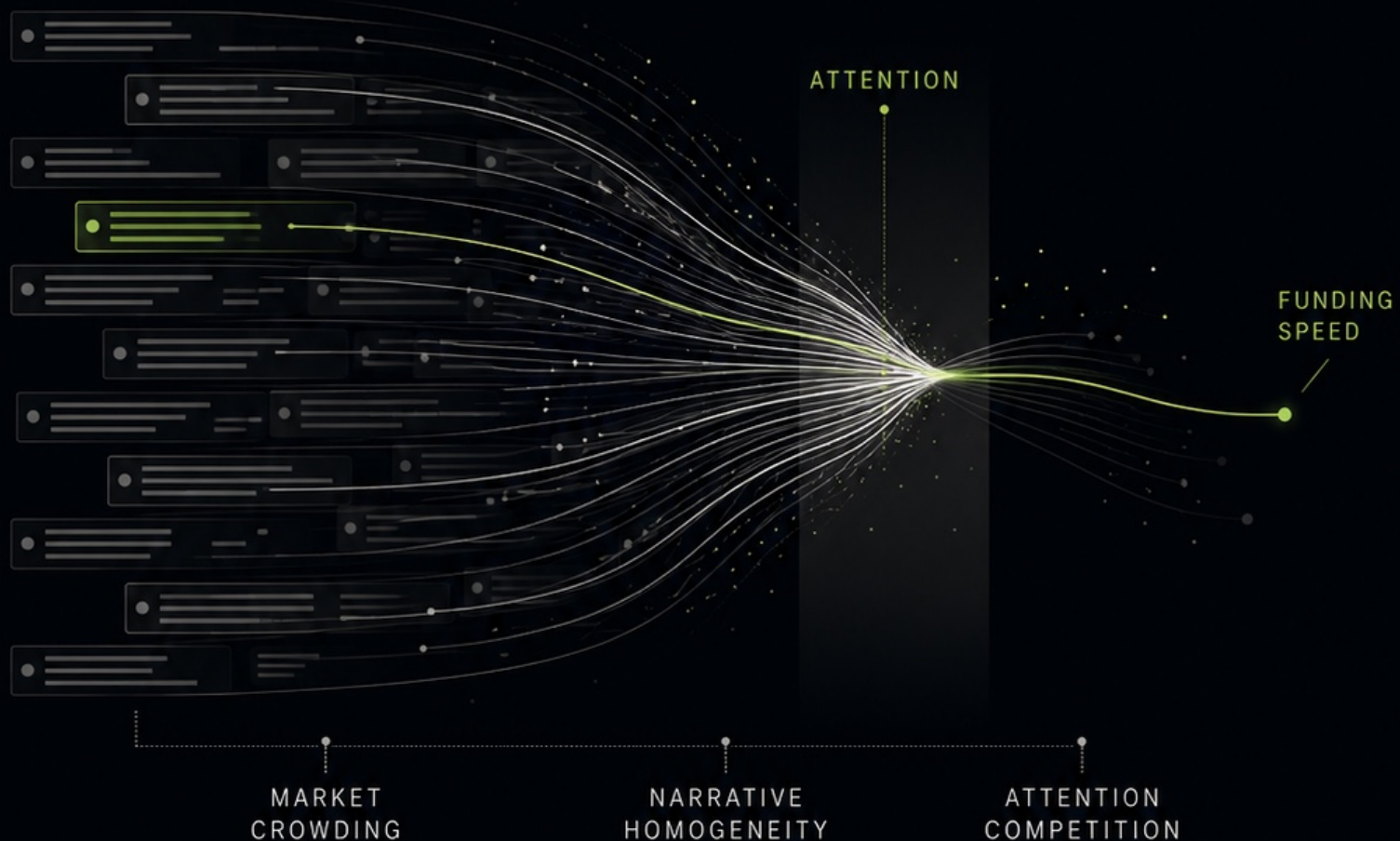


WHEN EVERY STORY SOUNDS THE SAME

Separating Market Crowding, Active Narrative Competition, and Narrative Wear-Out on Kiva



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Project Aim and Research Questions

Kiva lenders face an attention-constrained marketplace. Slow funding may reflect many concurrent projects, highly similar current alternatives, or recent repetition; these mechanisms require distinct interventions yet are often conflated.

Prior Kiva research finds competition slows funding, especially among close substitutes (Ly & Mason, 2012), while narrative distinctiveness is positively associated with crowdfunding outcomes (Williamson et al., 2021; Wang, 2024).

Repetition may matter only while similar loans remain available or persist after they leave. Advertising research suggests repeated exposure can reduce later attention and response (Pieters et al., 1999; Chae et al., 2019); its market-level analogue in prosocial crowdfunding remains untested.

We ask if slower funding is associated with similar current listings, recent repetition after those listings leave, or both.

Four research questions follow:

RQ1 – Active narrative saturation.

Is funding slower when the current same-sector choice set is both crowded and narratively homogeneous? Saturation is the interaction of volume and homogeneity; neither alone is sufficient.

RQ2 – Competition versus wear-out.

Beyond loans still fundraising, does saturation among completed loans that have left the page remain associated with slower funding? We call this market-level wear-out and test whether it survives removing repeated institutional boilerplate.

RQ3 – Evolution of the two channels.

How do the active and recent saturation channels change from 2016 to 2025? We report year-specific marginal effects rather than assume a single linear trend or attribute changes to one event.

RQ4 – Segment boundaries.

Where are these patterns strongest across sectors and countries, and does the relevant attention market operate mainly within sectors or across the whole platform?

Proposed Analytical Approaches

Outcome and comparison pools

Each loan is analysed separately. Funding time is measured in hours from listing to full funding and modelled as $\log(1 + \text{hours})$ to limit the influence of unusually slow loans. If some loans are still unfunded when observation ends, we keep them in the analysis using a censored accelerated failure-time (AFT) model rather than dropping them or assuming a raisedDate.

For each loan, we define its active pool as other same-sector loans that were already available for funding when it was posted. A loan remains in the pool from its fundraisingDate

until its raisedDate, or until an assumed closing date if it was not fully funded. This pool represents the alternatives a lender could have chosen from at that moment. Because slower loans stay available for longer, they appear in more later pools, so the pool's size and composition can partly depend on previous funding outcomes.

We also create a 14-day comparison pool of other same-sector loans posted during the previous 14 days, regardless of their later funding outcomes. Unlike the active pool, its composition is not affected by how quickly those loans were funded. In the 100-loan sample, 75% of loans fund within 15 days, supporting 14 days as an initial window. Once the full dataset is available, we will recalculate the funding-time distribution and adjust the window if needed.

At the beginning, check full-data loan statuses. If end dates for non-funded loans are missing, we assess whether they can be reconstructed reliably; if not, the time-based pool becomes primary. All rules are fixed before modelling.

Homogeneity estimates are less reliable when a pool contains very few loans. We therefore require at least 10 loans in the main analysis and repeat the analysis with minimum pool sizes of 5 and 20 as sensitivity checks. Smaller and empty pools are reported separately rather than treated as zero.

Measures and primary model

We measure each loan's narrative environment in two windows: current fundable same-sector loans and an earlier lagged pool. In each window, volume captures how many loans are present, homogeneity how similar their stories are, and their interaction represents saturation. Homogeneity uses mean cosine similarity, so it does not automatically rise with pool size.

C, H, V, and G are standardised once using 2016–2024 data before $C \times H$ and $V \times G$ are created. The same scaling is then applied to the 2025 holdout.

Measure	Current pool	Lagged pool
Volume	C_i log current same-sector loan count	V_i time-decayed earlier count
Homogeneity	H_i mean similarity to current loans	G_i time-decayed mean similarity to earlier stories
Saturation	$C_i \times H_i$	$V_i \times G_i$

To separate earlier repetition from current competition, we create a 30-day lagged window in which most earlier loans have finished fundraising before the focal loan is posted. This window begins at the 95th percentile of the fundraising period. Stories closer to the focal loan receive greater weight, using a seven-day half-life. We also repeat the analysis after removing any loans from this earlier window that were still fundraising when the focal loan was posted.

$$\log(1 + \text{FundingHours}_i) = \beta_0 + \beta_1 C_i + \beta_2 H_i + \beta_3 (C_i \times H_i) + \beta_4 V_i + \beta_5 G_i + \beta_6 (V_i \times G_i) + \gamma X_i + \text{Country FE} + \text{Activity FE} + \text{Week FE} + \varepsilon_i.$$

The model relates the six narrative measures to funding duration while controlling for loan, borrower, repayment and posting-time characteristics, platform-wide volume, and country, activity and week fixed effects (FE). These FE account for persistent differences across countries and activities and common changes affecting loans posted in the same week. RQ1 tests C×H; RQ2 jointly tests V, G and V×G. Results are interpreted as conditional associations, not causal effects.

Text representation and template decomposition

Kiva's browse view shows the short use statement, while the full description requires click-through. Saturation operates at the scan stage, which points to use, but short text makes similarity noisier, which points to description.

At the beginning, we compute homogeneity on both and report each field's similarity distribution, zero-overlap share, and post-masking token count, fixing the primary field before any outcome is viewed. whySpecial is secondary, entering only if Day 1 confirms coverage and variance.

Text is cleaned and masked, then represented as L2-normalised word unigram and bigram TF-IDF vectors. A mask replaces borrower names, locations, currencies and amounts, institutional abbreviations, dates, etc., with placeholders. This increases similarity, but H focuses more on estimating repetition caused by narrative framing. TF-IDF captures lexical, not semantic, repetition, so it understates homogeneity.

Profiles are uploaded by Lending Partners and may be edited or translated (Kiva, n.d.), so repetition may reflect institutional templates rather than borrowers' choices. During data processing, we identified high-frequency word-for-word repeating sequences as templates. Then, we estimated each headline model on both the raw and the residual text before and after template removal, while retaining boilerplate share as a new variable. The comparison of the two results serves as the basis for distinguishing between institutional repetition and substantive narrative overlap.

Heterogeneity Analysis

We allow the funding-time associations of current and recent saturation to vary by fundraising year, sector, and country through interactions.

For RQ3, we interact both saturation terms with 2016–2024 year indicators, using 2016 as the reference. Interactions show how each later-year association differs from 2016. As the period spans COVID-19 and platform changes, these differences indicate temporal heterogeneity, not any single event's effect.

$$\begin{aligned}\log(1 + \text{FundingHours}_i) = & \beta_0 + \beta_1 C_i + \beta_2 H_i + \beta_3 (C_i \times H_i) \\ & + \beta_4 V_i + \beta_5 G_i + \beta_6 (V_i \times G_i) \\ & + \delta' [(C_i \times H_i) \times \text{Year}_i] \\ & + \theta' [(V_i \times G_i) \times \text{Year}_i] \\ & + \gamma X_i + \text{Country FE} + \text{Activity FE} + \text{Week FE} + \varepsilon_i.\end{aligned}$$

For RQ4, we interact both saturation terms with sector and country indicators.

$$\begin{aligned}
\log(1 + \text{FundingHours}_i) &= \beta_0 + \beta_1 C_i + \beta_2 H_i + \beta_3 (C_i \times H_i) \\
&+ \beta_4 V_i + \beta_5 G_i + \beta_6 (V_i \times G_i) \\
&+ \sum_{s \neq s_0} \alpha_s (C_i \times H_i) I(\text{Sector}_i = s) \\
&+ \sum_{s \neq s_0} \eta_s (V_i \times G_i) I(\text{Sector}_i = s) \\
&+ \gamma X_i + \text{Country FE} + \text{Activity FE} + \text{Week FE} + \varepsilon_i.
\end{aligned}$$

$$\begin{aligned}
\log(1 + \text{FundingHours}_i) &= \beta_0 + \beta_1 C_i + \beta_2 H_i + \beta_3 (C_i \times H_i) \\
&+ \beta_4 V_i + \beta_5 G_i + \beta_6 (V_i \times G_i) \\
&+ \sum_{c \neq c_0} \kappa_c (C_i \times H_i) I(\text{Country}_i = c) \\
&+ \sum_{c \neq c_0} \rho_c (V_i \times G_i) I(\text{Country}_i = c) \\
&+ \gamma X_i + \text{Country FE} + \text{Activity FE} + \text{Week FE} + \varepsilon_i.
\end{aligned}$$

Across these specifications, the interaction coefficients capture how the effects of current and recent saturation for a given year, sector, or country differ from those of the corresponding reference group. The group-specific effect is therefore given by the baseline saturation coefficient plus the relevant interaction coefficient.

Joint interaction tests assess whether the saturation effects differ overall across sectors or countries. Where heterogeneity is supported, we report group-specific marginal effects to show where current and recent saturation are stronger or weaker.

Data Items to Be Used

The official data contain 1,453,846 loan records from 1 January 2016 to 31 December 2025 (UNSW, 2026). Core fields are id, status, fundraisingDate, raisedDate, description, use, whySpecial, loanAmount, borrowerCount, gender, lenderRepaymentTerm, repaymentInterval, sector, activity, country_name, region, country_ppp, and disbursalDate.

Names, exact locations, and full descriptions will not appear in outputs; results are aggregated, and funding duration will not be read as borrower quality. Because every measure is taken at posting, fields resolved later — including fundsLentInCountry — are excluded.

Expected Outcomes and Managerial Relevance

The presentation contrasts active saturation, recent saturation beyond current competition, and variation across years, sectors, and countries.

The managerial output is a Narrative Attention Engine scoring each listing on current and recent crowding, homogeneity and saturation, and on boilerplate share; 2025 records are held back to check the scoring transfers. High current saturation suggests staggering substitutes or diversifying listing pages; high recent saturation, refreshing overused framing; high boilerplate, Lending Partner template support; and a distinctive loan still disadvantaged should receive exposure support, not repeated rewriting.

Because the two channels imply different remedies, the evolutionary result matters operationally: a platform whose bottleneck has shifted from competition toward wear-out needs editorial refresh, not queue management. Stress tests will report how many loans fall below a 72-hour predicted threshold under hypothetical profiles. These are not intervention forecasts; operational change requires prospective experiments.

Strategic contribution: not "make every borrower tell a louder story, "But help Kiva place each authentic story in a less crowded conversation without shifting the burden of platform competition onto borrowers.

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